

# *E9 205 Machine Learning for Signal Processing*

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**ML, MAP, MMSE and Gaussian  
Modeling**

12-09-2016

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# Recap ...

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- ❖ Decision Theory
  - ❖ Inference problem
    - ❖ Finding the joint density  $p(\mathbf{x}, \mathbf{t})$
  - ❖ Decision problem
    - ❖ Using the inference to make the classification or regression decision



# Decision Problem - Classification

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- ❖ Minimizing the mis-classification error
- ❖ Decision based on maximum posteriors

$$\operatorname{argmax}_j p(C_j|\mathbf{x})$$

- ❖ Loss matrix
  - ❖ Minimizing the expected loss

$$\operatorname{argmax}_j \sum_k L_{k,j} p(C_k|\mathbf{x})$$

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# Approaches for Inference and Decision

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I. Finding the joint density from the data.

$$p(C_k|\mathbf{x}) \propto p(\mathbf{x}|C_k)p(C_k)$$

II. Finding the posteriors directly.

III. Using discriminant functions for classification.



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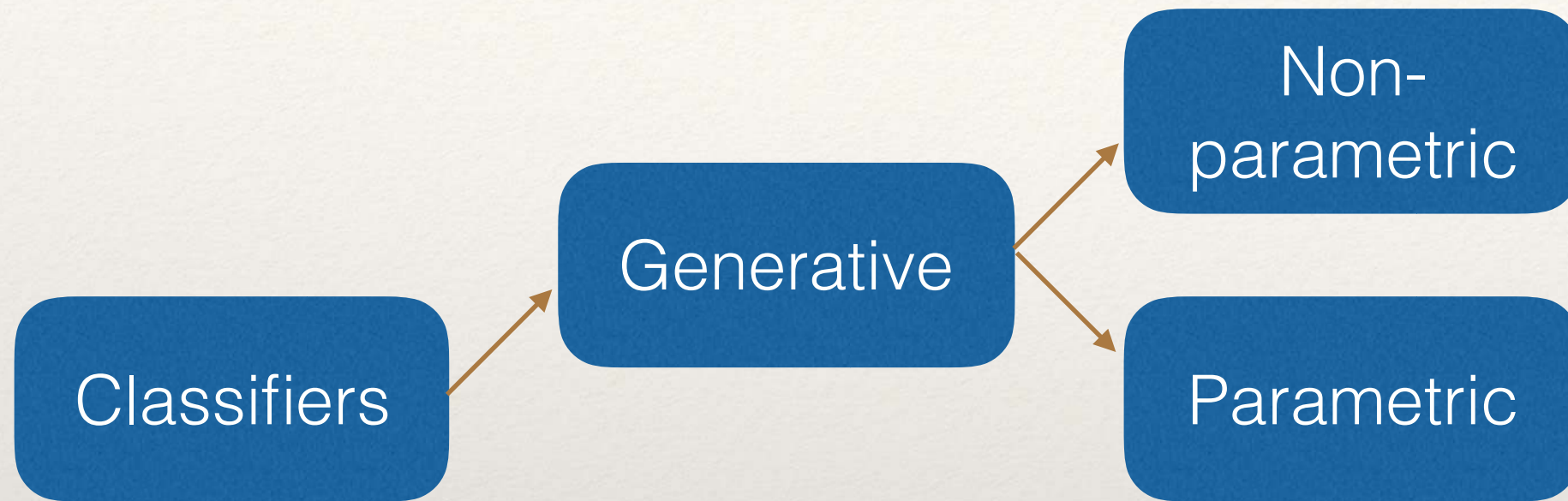
# Decision Rule for Regression

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- ❖ Minimum mean square error loss
- ❖ Solution is conditional expectation.

# Generative Modeling

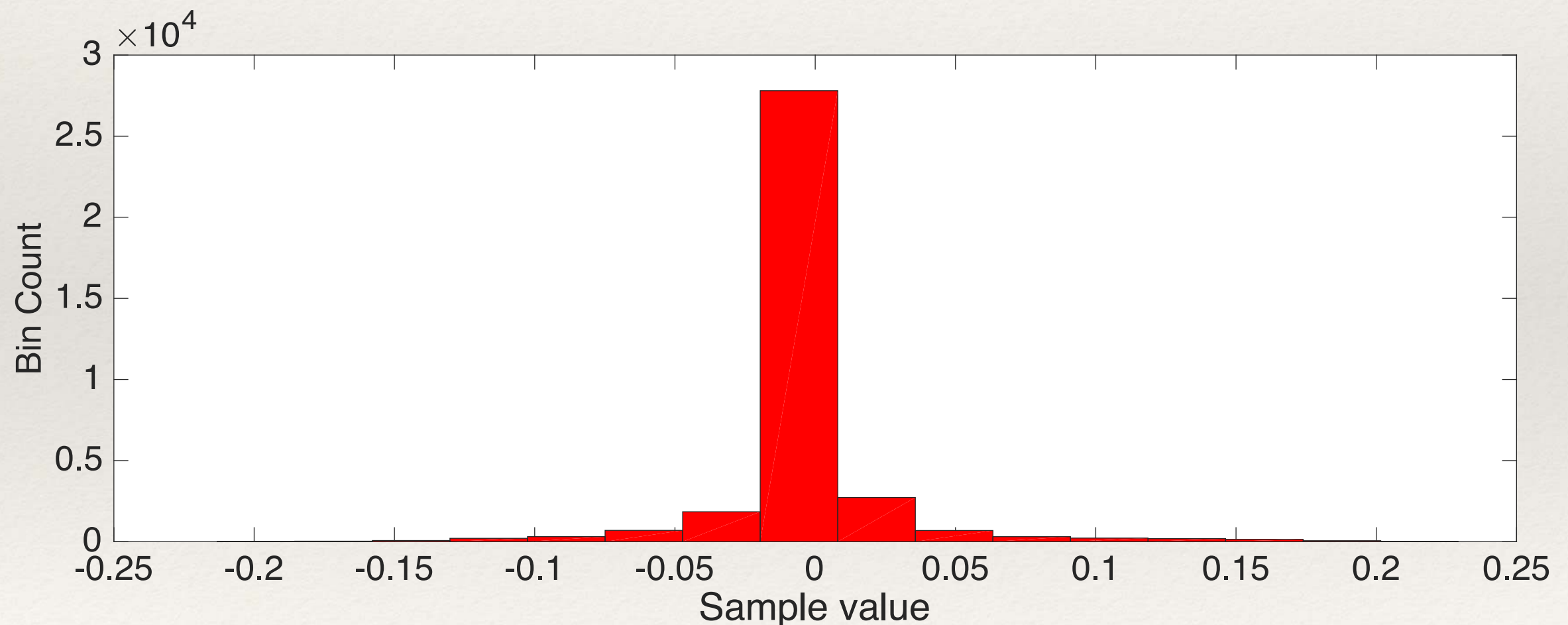
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# Non-parametric Modeling

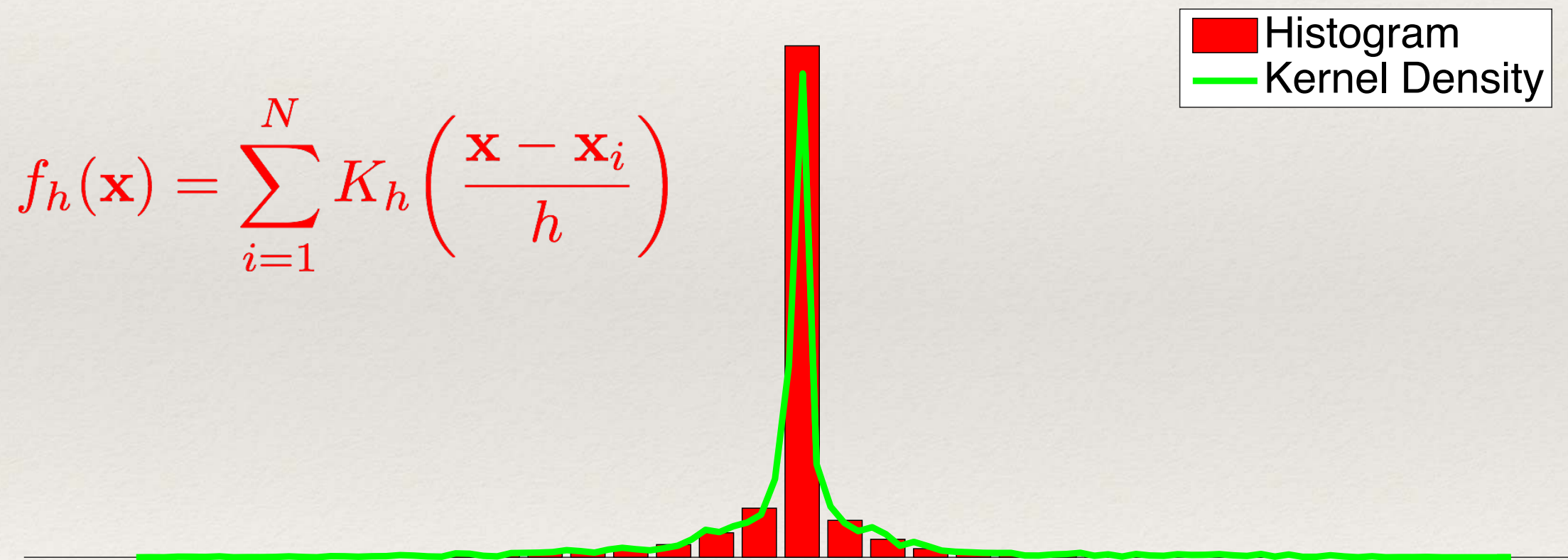
- **Non-parametric** models do not specify an apriori set of parameters to model the distribution. Example - Histogram



The density is not smooth and has block like shape.

# Non-parametric Modeling

- **Non-parametric** models do not specify an apriori set of parameters to model the distribution.
  - Example - Kernel Density Estimators



**Kernel** is a smooth function which obeys certain properties



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# Non-parametric Modeling

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- Non-parametric methods are dependent on number of data points
  - Estimation is difficult for large datasets.
- Likelihood computation and model comparisons are hard.
- Limited use in classifiers

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# Parametric Models

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- ❖ Collection of probability distributions which are described by a finite dimensional parameter set

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_K) \quad P = \{P_{\boldsymbol{\theta}}\}$$

- Examples -

- Poisson Distribution

$$p_{\lambda}(j) = \frac{\lambda^j}{j!} e^{-\lambda}$$

- Bernoulli Distribution

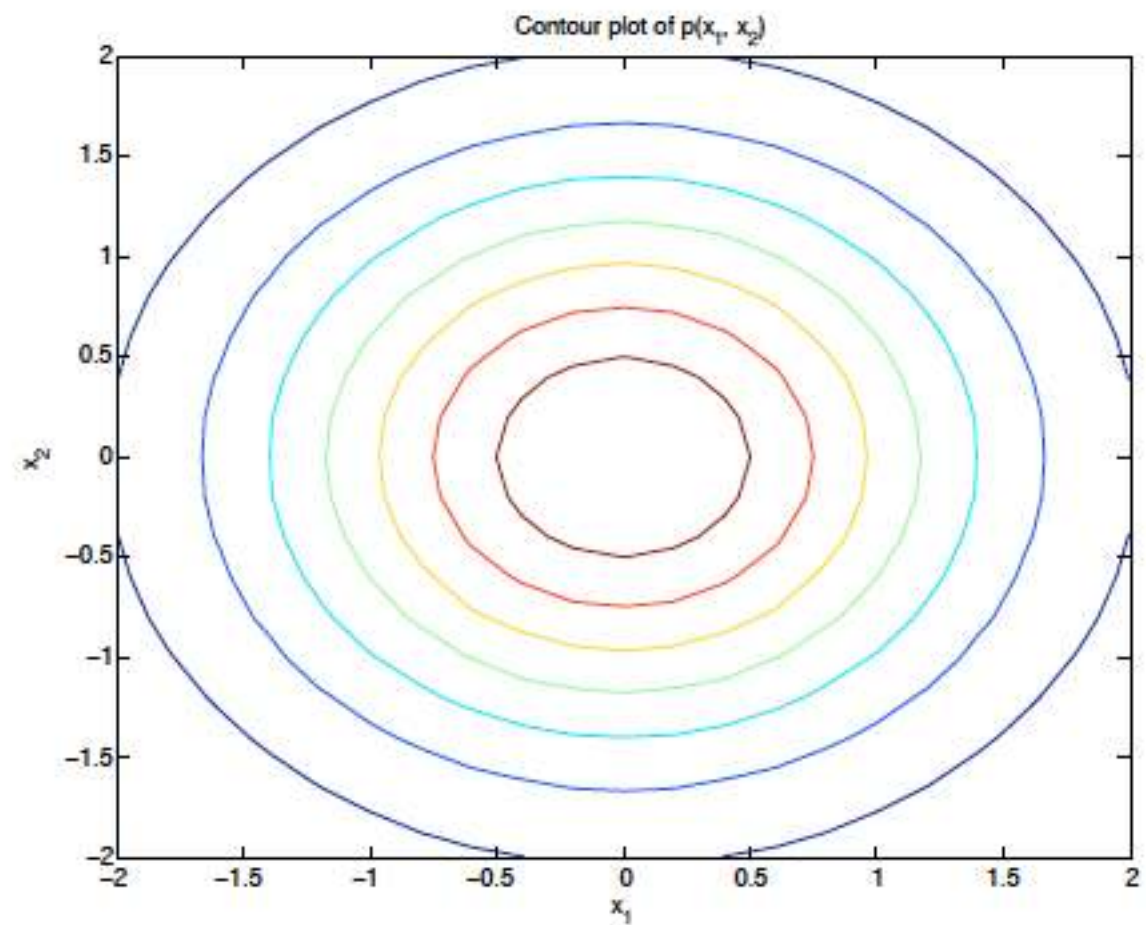
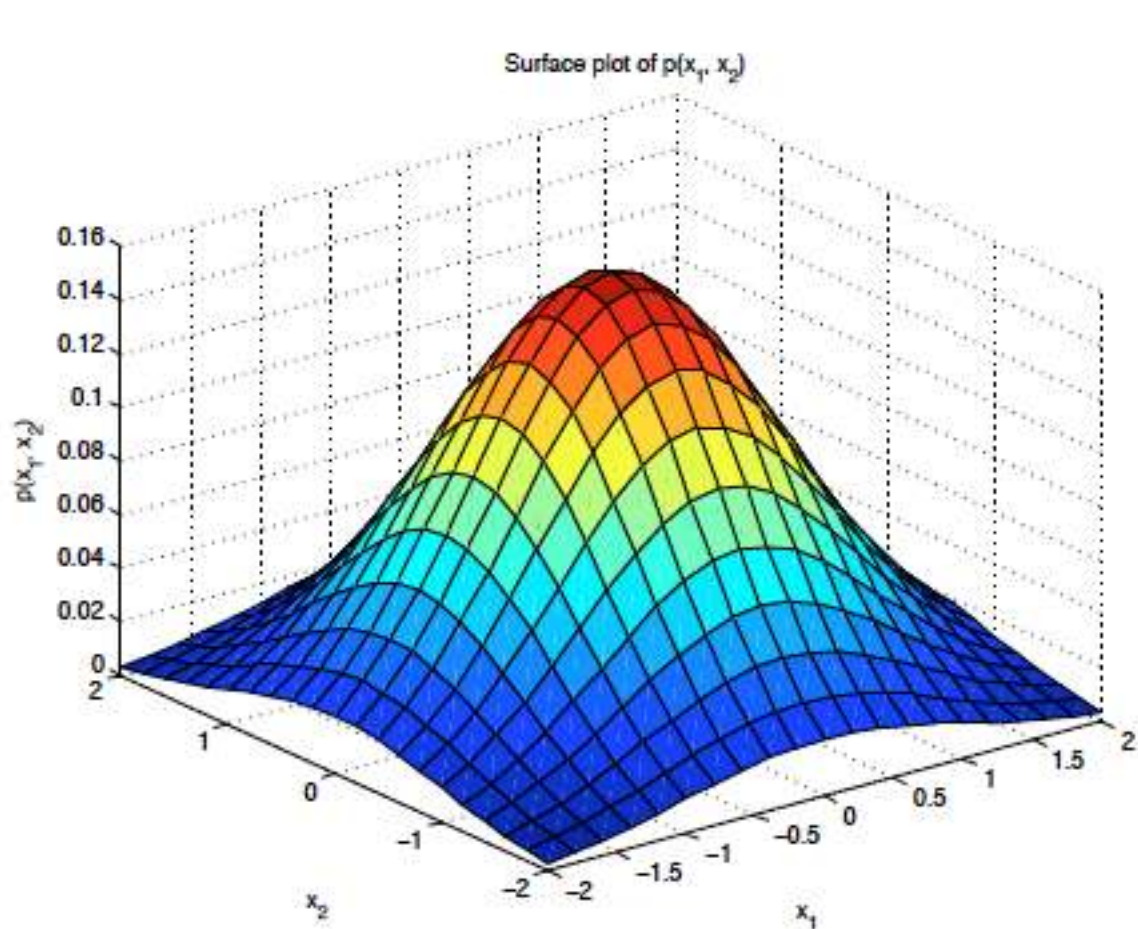
$$p(\mathbf{x}|\boldsymbol{\mu}) = \prod_{i=1}^D \mu_i^{x_i} (1 - \mu_i)^{1-x_i}$$

- Gaussian Distribution

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^* \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$



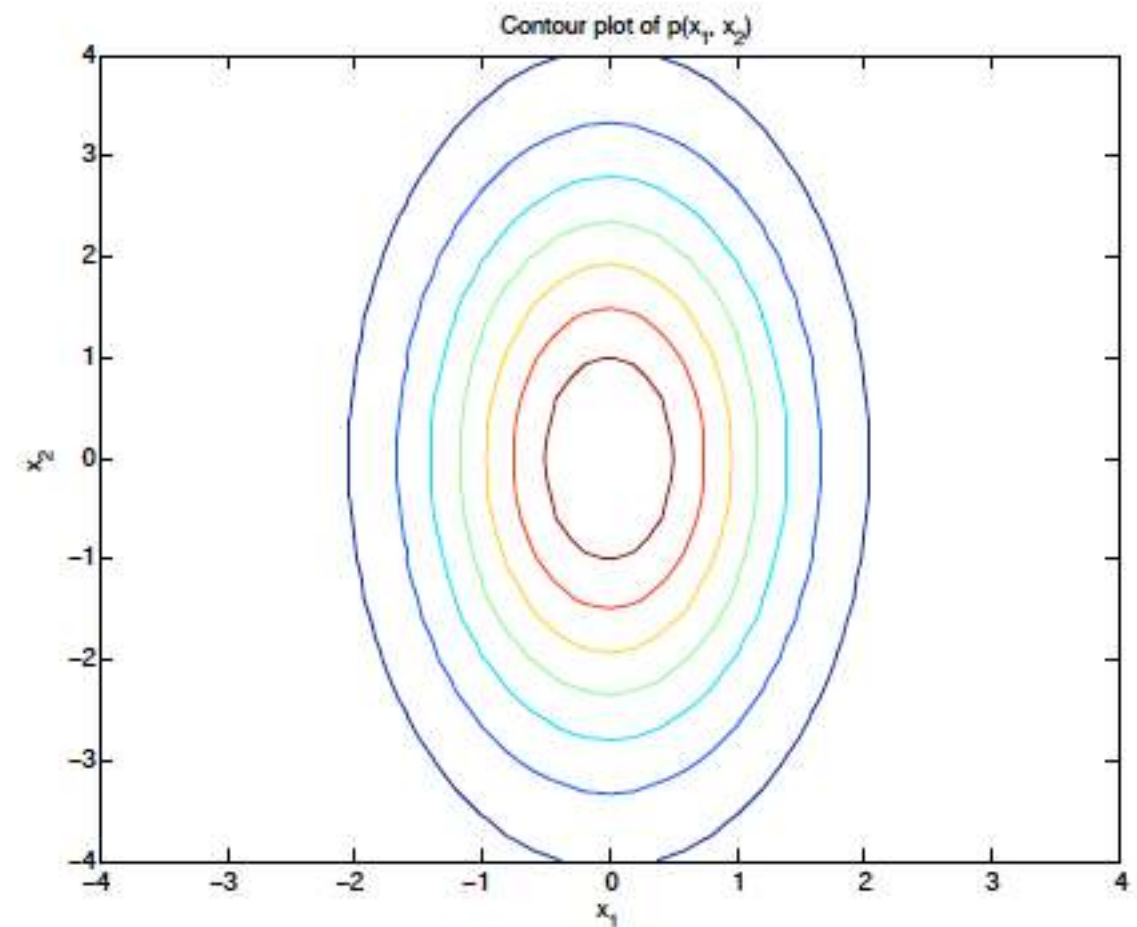
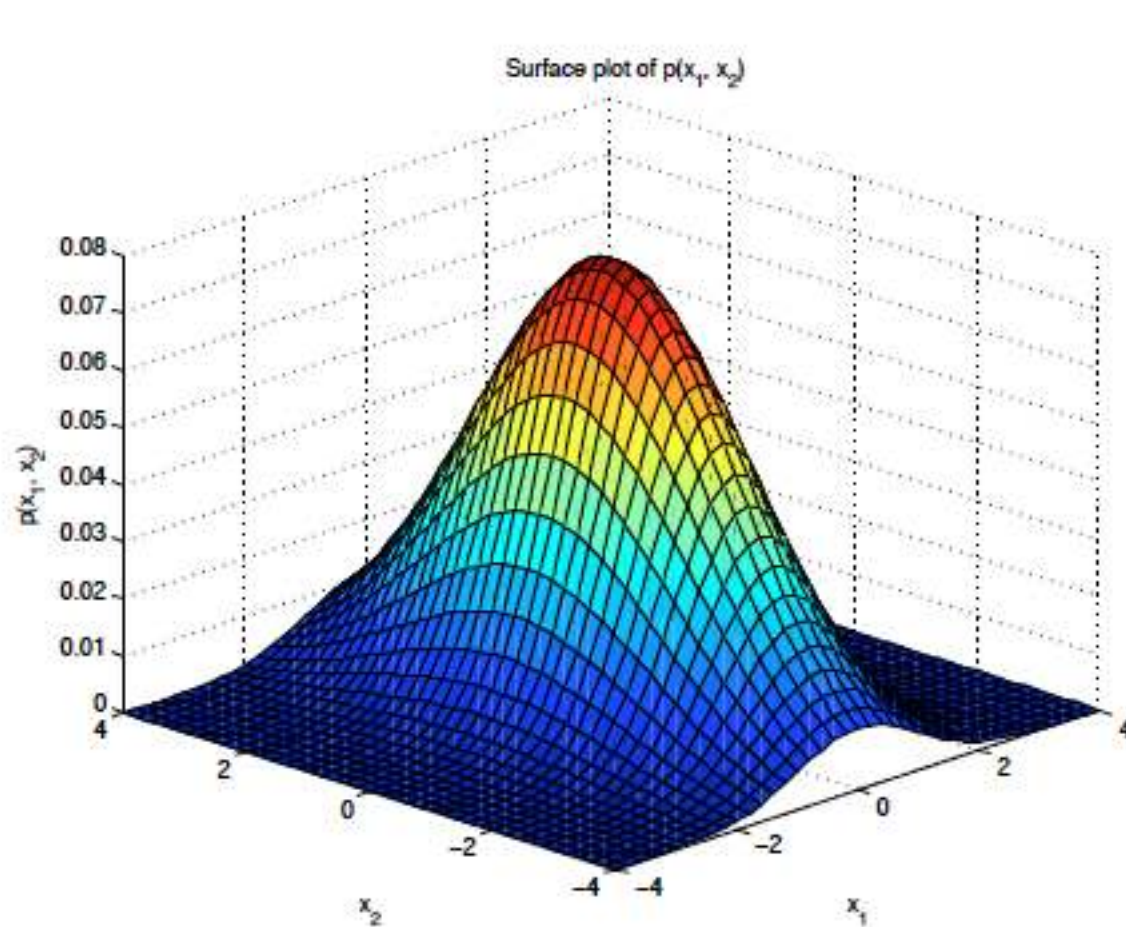
# Gaussian Distribution



Points of equal probability lie on on contour  
Diagonal Gaussian with Identical Variance



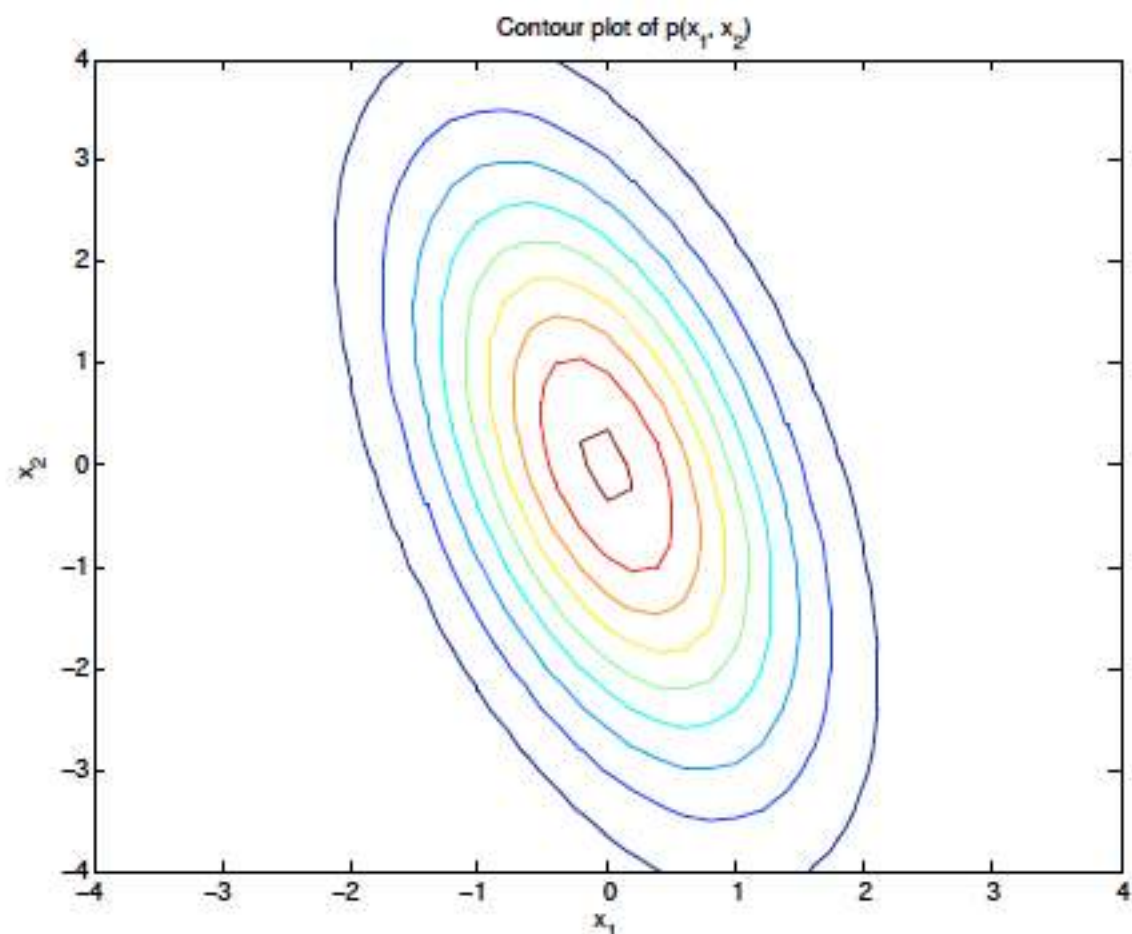
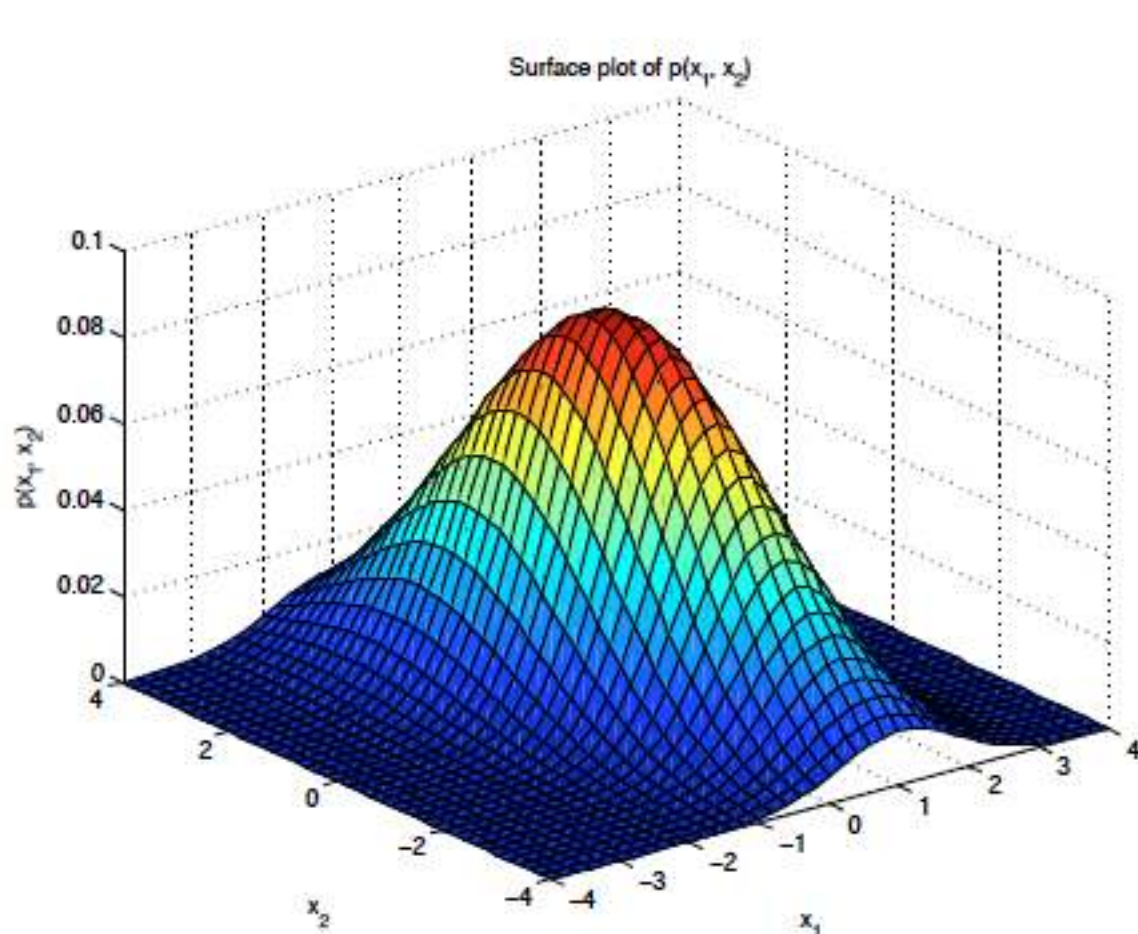
# Gaussian Distribution



Diagonal Gaussian with different variance

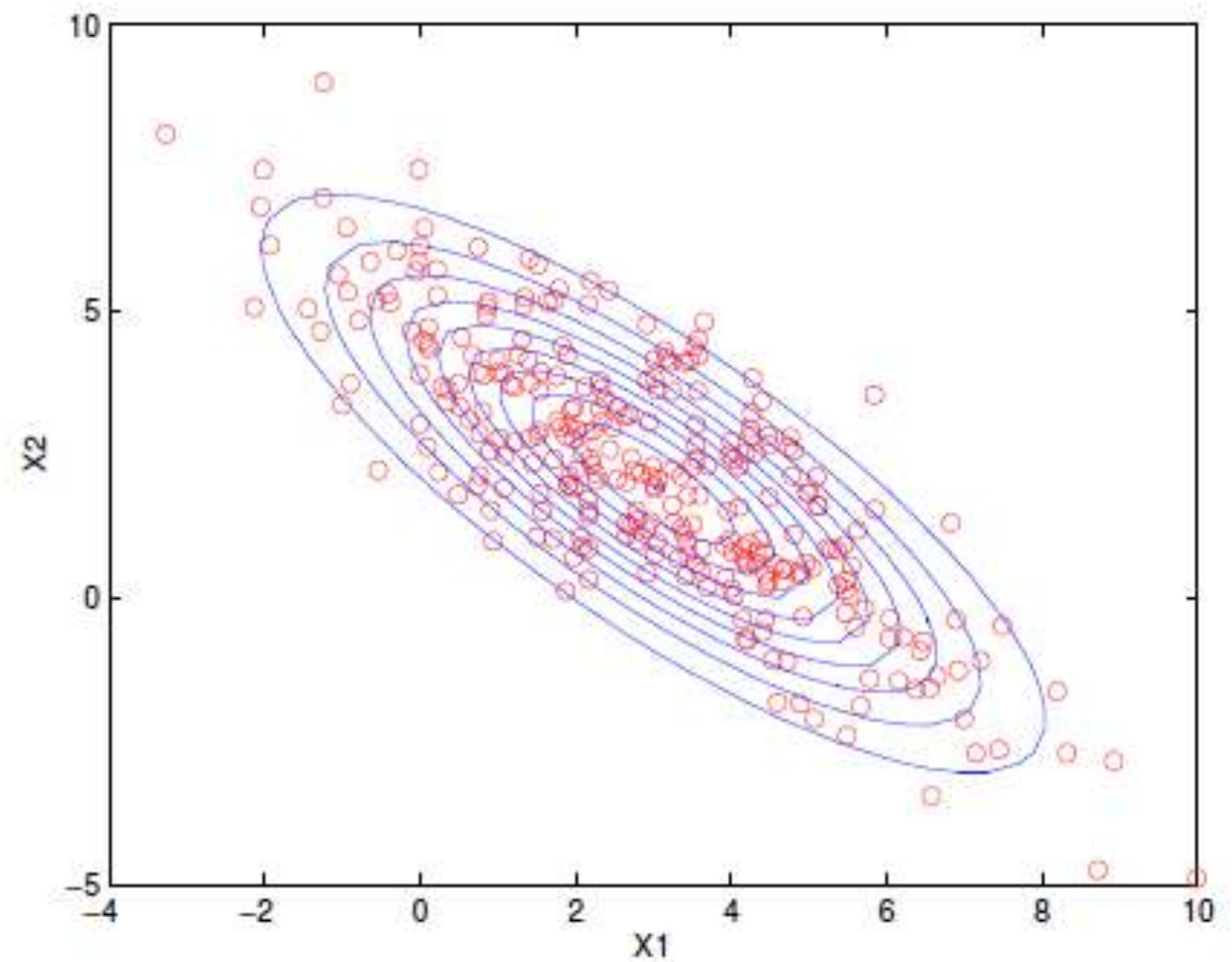
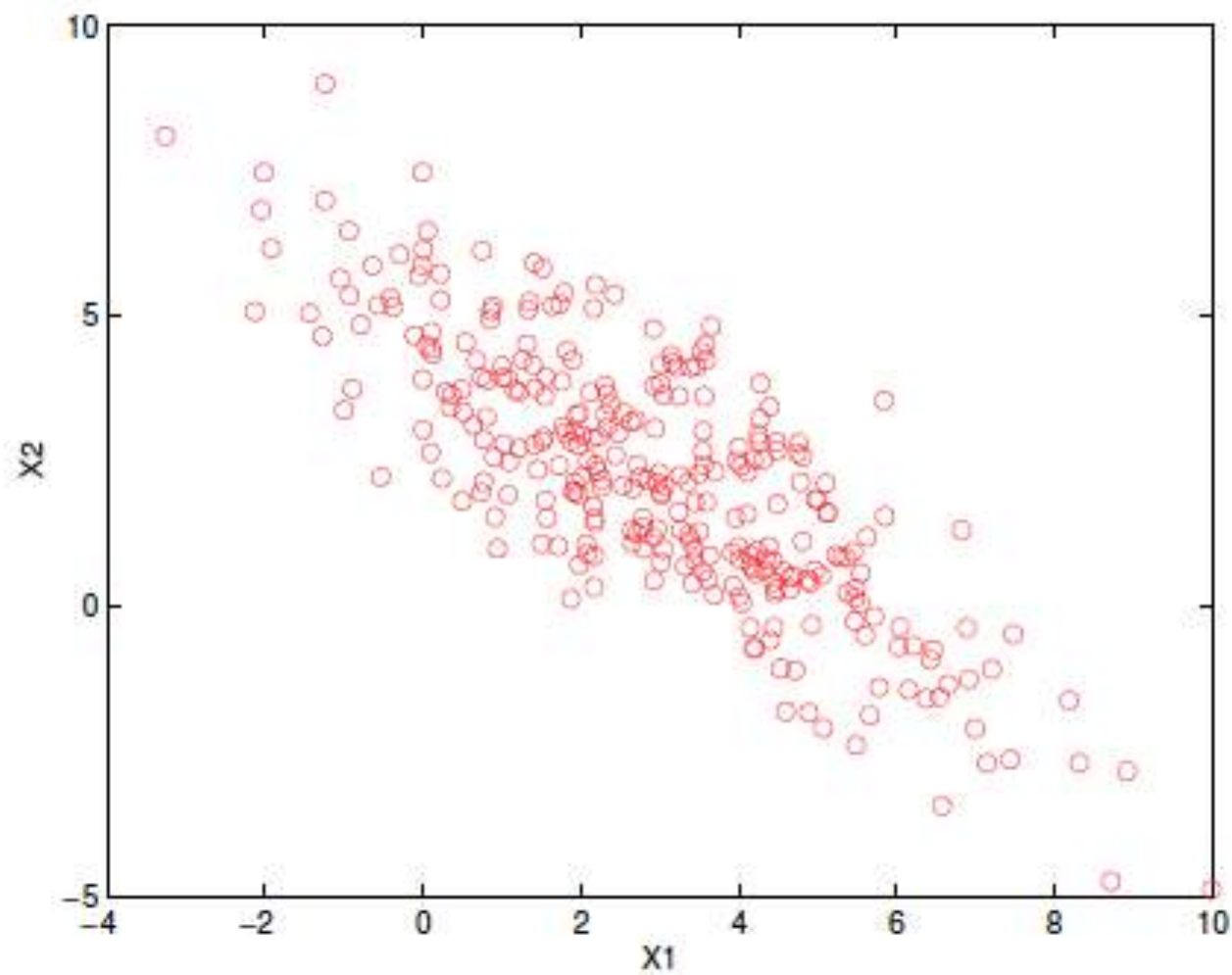


# Gaussian Distribution



Full covariance Gaussian distribution

# Gaussian Distribution





# Finding the parameters of the Model

- ❖ The Gaussian model has the following parameters

$$\theta = (\mu, \Sigma)$$

- ❖ Total number of parameters to be learned for D dimensional data is  $D^2 + D$
- ❖ Given N data points  $\{\mathbf{x}_i\}_{i=1}^N$  how do we estimate the parameters of model.
  - ❖ Several criteria can be used
  - ❖ The most popular method is the maximum likelihood estimation (MLE).

# MLE

Define the likelihood function as  $L(\boldsymbol{\theta}) = \prod_{i=1}^N p(\mathbf{x}_i | \boldsymbol{\theta})$

The maximum likelihood estimator (MLE) is

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} L(\boldsymbol{\theta})$$

The MLE satisfies nice properties like

- Consistency (convergence to true value)
- Efficiency (has the least Mean squared error).



# MLE

For the Gaussian distribution

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^* \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

$$L(\boldsymbol{\theta}) = \prod_{i=1}^N p(\mathbf{x}_i|\boldsymbol{\theta})$$

$$\log L(\boldsymbol{\theta}) = -\frac{ND}{2} - \frac{N}{2} \log |\boldsymbol{\Sigma}| - \frac{1}{2} \sum_{i=1}^N \left( (\mathbf{x}_i - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) \right)$$

To estimate the parameters  $\frac{\partial \log L}{\partial \boldsymbol{\mu}} = 0$